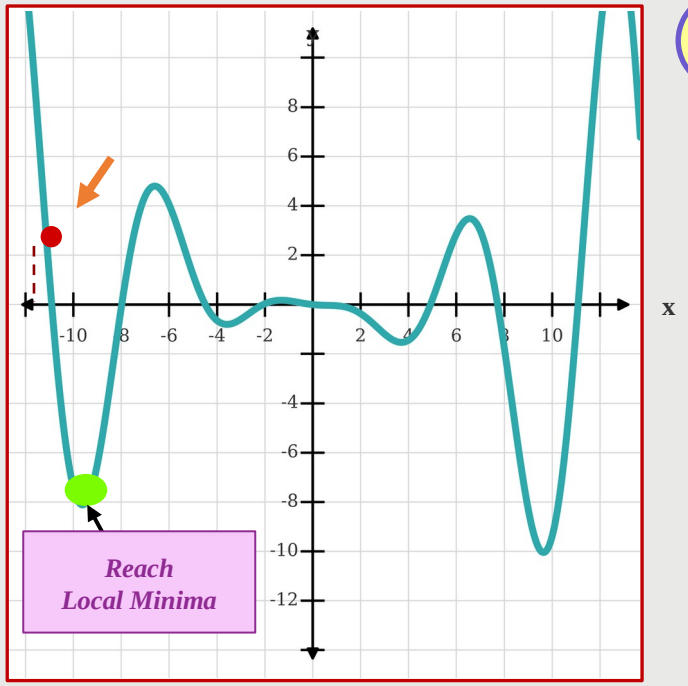
# Non-Convex Function

## May reach different minimum value based on:

## initial point and learning rate

$$f(x) = \frac{x^2\cos(x) - x}{10}$$




| step | 0.1 |
|---|---|
| initial X | -11 |
| delta | 0.1 |

***X*** is in between -9.7 and color red bold -9.6

***f(x)*** is in between -8.08 and -8.011

| x | -11 | -10.9 | -10.8 | -10.7 | -10.6 | -10.5 | -10.4 | -10.3 | -10.2 | -10.1 | -10 | -9.9 | -9.8 | -9.7 | -9.6 | -9.7 | -9.6 | -9.7 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| f(x) | 1.15 | -0.04 | -1.19 | -2.26 | -3.27 | -4.19 | -5.03 | -5.77 | -6.41 | -6.95 | -7.39 | -7.72 | -7.96 | -8.08 | -8.11 | -8.08 | -8.11 | -8.08 |
| x-delta | -11.1 | -11 | -10.9 | -10.8 | -10.7 | -10.6 | -10.5 | -10.4 | -10.3 | -10.2 | -10.1 | -10 | -9.9 | -9.8 | -9.7 | -9.8 | -9.7 | -9.8 |
| f(x-delta) | 2.39 | 1.15 | -0.04 | -1.19 | -2.26 | -3.27 | -4.19 | -5.03 | -5.77 | -6.41 | -6.95 | -7.39 | -7.72 | -7.96 | -8.08 | -7.96 | -8.08 | -7.96 |
| x+delta | -10.9 | -10.8 | -10.7 | -10.6 | -10.5 | -10.4 | -10.3 | -10.2 | -10.1 | -10 | -9.9 | -9.8 | -9.7 | -9.6 | -9.5 | -9.6 | -9.5 | -9.6 |
| f(x+delta) | -0.04 | -1.19 | -2.26 | -3.27 | -4.19 | -5.03 | -5.77 | -6.41 | -6.95 | -7.39 | -7.72 | -7.96 | -8.08 | -8.11 | -8.05 | -8.11 | -8.05 | -8.11 |
| slope | -24.3 | -23.4 | -22.2 | -20.8 | -19.3 | -17.6 | -15.8 | -13.8 | -11.8 | -9.8 | -7.7 | -5.7 | -3.6 | -1.5 | 0.3 | -1.5 | 0.3 | -1.5 |
| next x | -10.9 | -10.8 | -10.7 | -10.6 | -10.5 | -10.4 | -10.3 | -10.2 | -10.1 | -10 | -9.9 | -9.8 | -9.7 | -9.6 | -9.7 | -9.6 | -9.7 | -9.6 |